Curve Sketching Practice

1. Consider the function $f(x) = \frac{8x}{(x-1)^2}$. Sketch the graph of the function using the provided information. Be sure to clearly label all asymptotes, critical points, extrema, and points of inflection on your graph.

Domain and Asymptotes:

- The domain of $f(x)$ is $(-\infty, 1) \cup (1, \infty)$.
- Vertical asymptote at $x = 1$.
- Horizontal asymptote at $y = 0$ (as $x \to \infty$ and $x \to -\infty$).

First Derivative (Critical Points & Increasing/Decreasing):

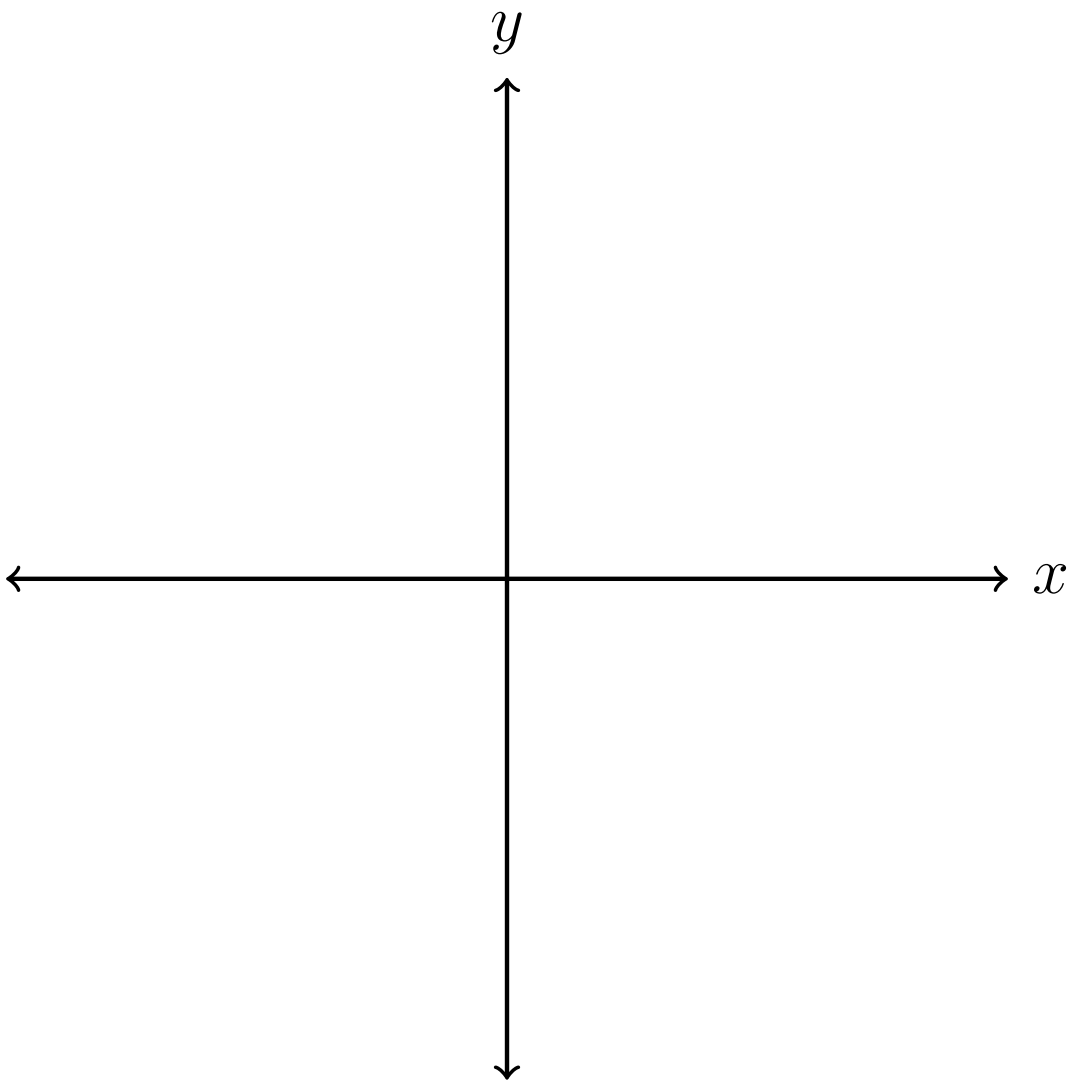
- $f'(x) = \dfrac{-8(x+1)}{(x-1)^3}$
- Critical point at $x = -1$, where $f'(-1) = 0$. *(Note: The derivative $f'(x)$ does not exist at $x = 1$, but since $x = 1$ is not in the domain, it is a vertical asymptote rather than a critical point).*
- $f'(x) > 0$ on $(-1, 1)$.
- $f'(x) < 0$ on $(-\infty, -1) \cup (1, \infty)$.
- Local minimum at $x = -1$.
- No local maximum.

Second Derivative (Concavity & Inflection):

- $f''(x) = \dfrac{16(x+2)}{(x-1)^4}$
- $f''(x) > 0$ on $(-2, 1) \cup (1, \infty)$.
- $f''(x) < 0$ on $(-\infty, -2)$.
- Point of inflection at $x = -2$.

Points on the Graph (Function Values):

- $f(0) = 0$ (x and y-intercept).
- $f(-1) = -2$ (local minimum).
- $f(-2) = -\frac{16}{9} \approx -1.78$ (point of inflection).


y
x

2. Consider the function $f(x) = (x-2)^{2/3}$. Sketch the graph of the function using the provided information.

Domain and Asymptotes:

- The domain of $f(x)$ is $(-\infty, \infty)$.
- No vertical or horizontal asymptotes.

First Derivative (Critical Points & Increasing/Decreasing):

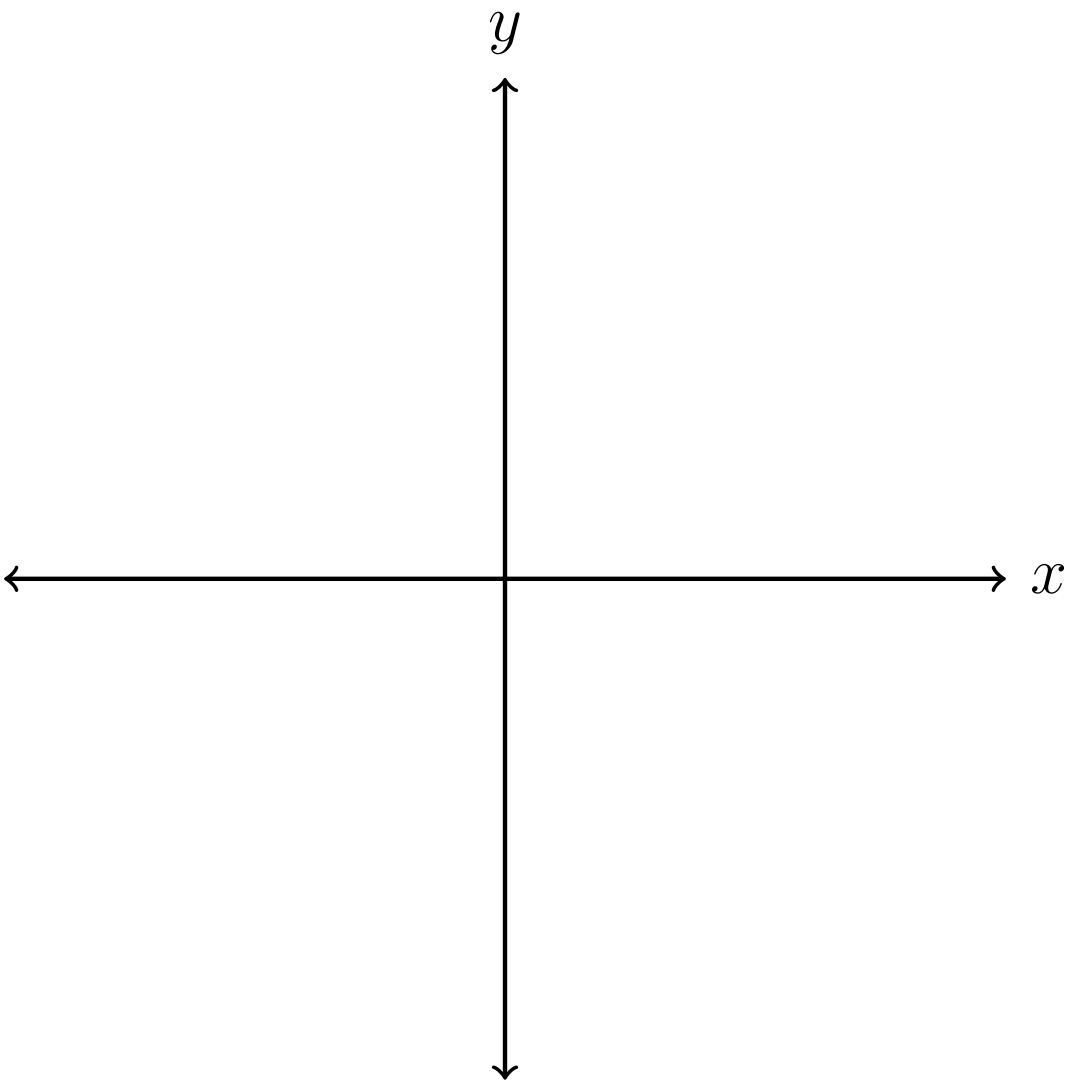
- $f'(x) = \dfrac{2}{3(x-2)^{1/3}}$
- Critical point at $x = 2$, where $f'(2)$ does not exist.
- $f'(x) > 0$ on $(2, \infty)$.
- $f'(x) < 0$ on $(-\infty, 2)$.
- Local minimum at $x = 2$.
- No local maximum.

Second Derivative (Concavity & Inflection):

- $f''(x) = \dfrac{-2}{9(x-2)^{4/3}}$
- $f''(x) < 0$ on $(-\infty, 2) \cup (2, \infty)$.
- The graph is concave down everywhere except at $x = 2$.
- No points of inflection.

Points on the Graph (Function Values):

- $f(2) = 0$ (local minimum and x-intercept).
- $f(0) = 2^{2/3} \approx 1.59$ (y-intercept).
- $f(1) = 1$.
- $f(3) = 1$.


y
x

3. Consider the rational function $f(x) = \frac{x^2+1}{x}$. Sketch the graph of the function using the provided information.

Domain and Asymptotes:

- The domain of $f(x)$ is $(-\infty, 0) \cup (0, \infty)$.
- Vertical asymptote at $x = 0$.
- Slant (oblique) asymptote at $y = x$.

First Derivative (Critical Points & Increasing/Decreasing):

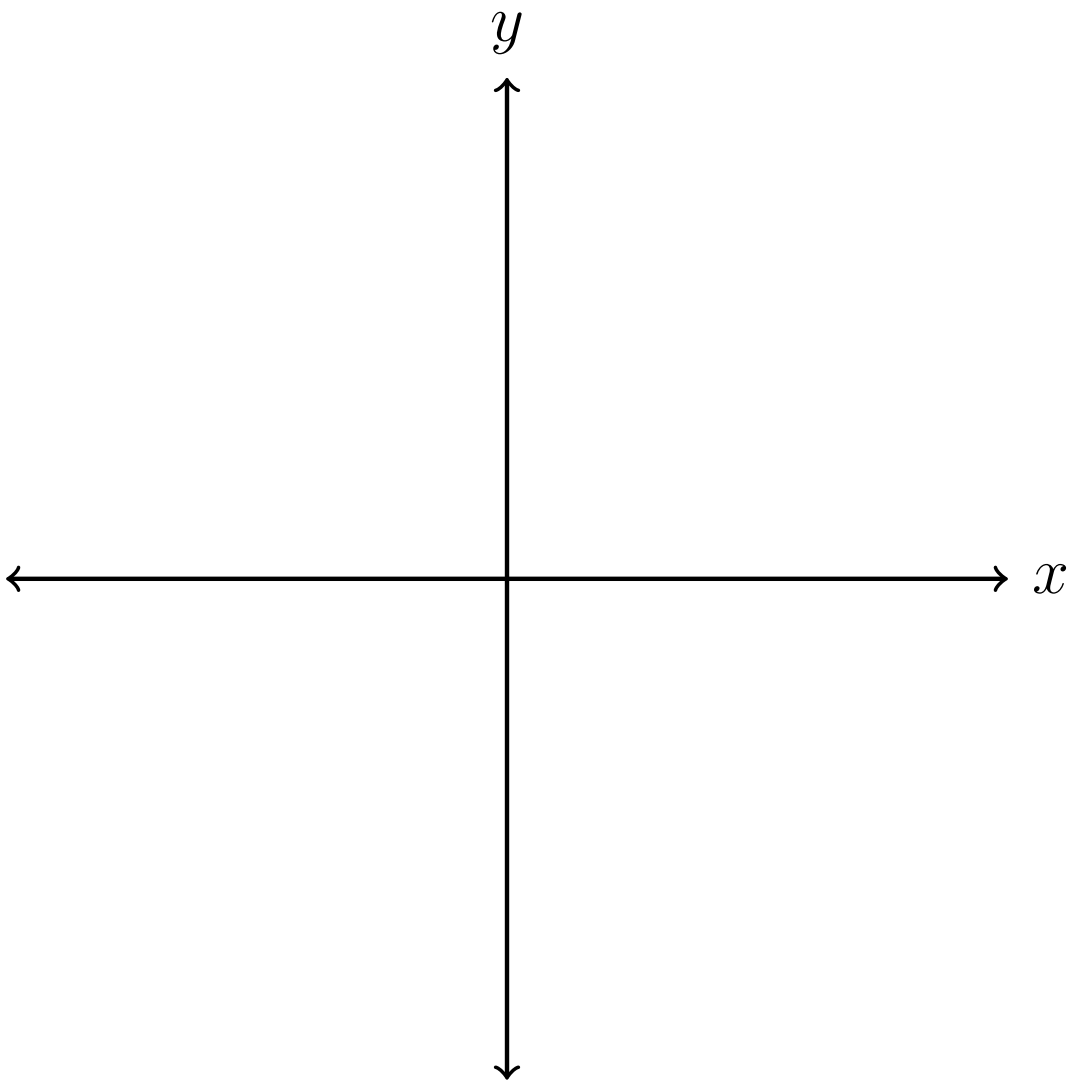
- $f'(x) = \dfrac{x^2 - 1}{x^2}$
- Critical points at $x = -1$ and $x = 1$, where $f'(x) = 0$. *(Note: $f'(0)$ does not exist, but $x = 0$ is a vertical asymptote).*
- $f'(x) > 0$ on $(-\infty, -1) \cup (1, \infty)$.
- $f'(x) < 0$ on $(-1, 0) \cup (0, 1)$.
- Local maximum at $x = -1$.
- Local minimum at $x = 1$.

Second Derivative (Concavity & Inflection):

- $f''(x) = \dfrac{2}{x^3}$
- $f''(x) > 0$ on $(0, \infty)$.
- $f''(x) < 0$ on $(-\infty, 0)$.
- No points of inflection (concavity changes across the vertical asymptote).

Points on the Graph (Function Values):

- $f(1) = 2$ (local minimum).
- $f(-1) = -2$ (local maximum).


y
x

4. Consider the trigonometric function $f(x) = \sin(x) + \cos(x)$ strictly on the interval $[0, 2\pi]$. Sketch the graph using the provided information.

Domain and Asymptotes:

- The domain is restricted to $[0, 2\pi]$.
- No asymptotes.

First Derivative (Critical Points & Increasing/Decreasing):

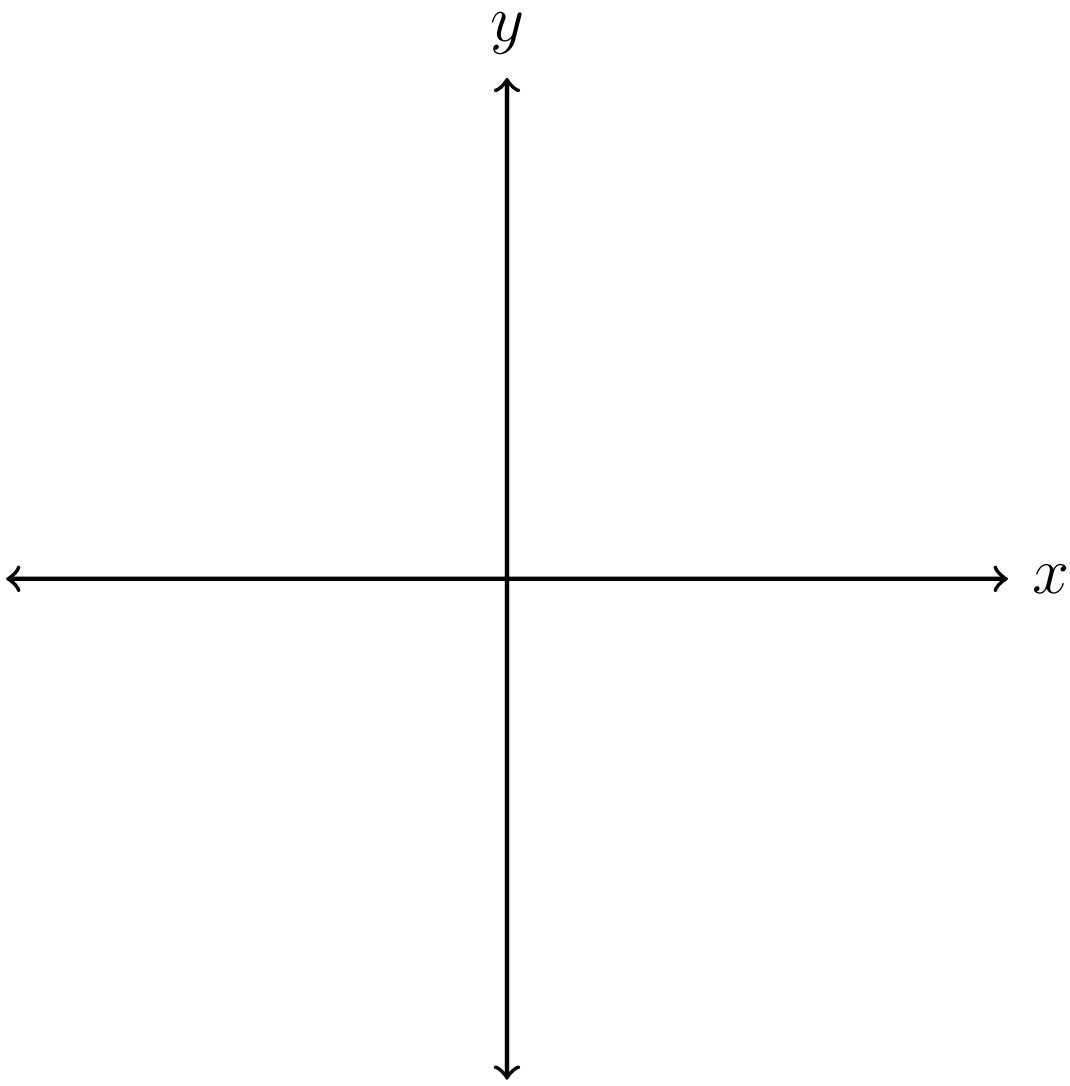
- $f'(x) = \cos(x) - \sin(x)$
- Critical points at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$, where $f'(x) = 0$.
- $f'(x) > 0$ on $(0, \frac{\pi}{4}) \cup (\frac{5\pi}{4}, 2\pi)$.
- $f'(x) < 0$ on $(\frac{\pi}{4}, \frac{5\pi}{4})$.
- Local maximum at $x = \frac{\pi}{4}$.
- Local minimum at $x = \frac{5\pi}{4}$.

Second Derivative (Concavity & Inflection):

- $f''(x) = -\sin(x) - \cos(x)$
- $f''(x) > 0$ on $(\frac{3\pi}{4}, \frac{7\pi}{4})$.
- $f''(x) < 0$ on $(0, \frac{3\pi}{4}) \cup (\frac{7\pi}{4}, 2\pi)$.
- Points of inflection at $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$.

Points on the Graph (Function Values):

- $f(0) = 1$ (starting endpoint).
- $f(2\pi) = 1$ (ending endpoint).
- $f(\frac{\pi}{4}) = \sqrt{2} \approx 1.41$ (local maximum).
- $f(\frac{5\pi}{4}) = -\sqrt{2} \approx -1.41$ (local minimum).
- $f(\frac{3\pi}{4}) = 0$ (point of inflection & x-intercept).
- $f(\frac{7\pi}{4}) = 0$ (point of inflection & x-intercept).


y
x

5. Consider the polynomial function $f(x) = x^4 - 4x^3 + 10$. Sketch the graph using the provided information.

Domain and Asymptotes:

- The domain of $f(x)$ is $(-\infty, \infty)$.
- No vertical or horizontal asymptotes.

First Derivative (Critical Points & Increasing/Decreasing):

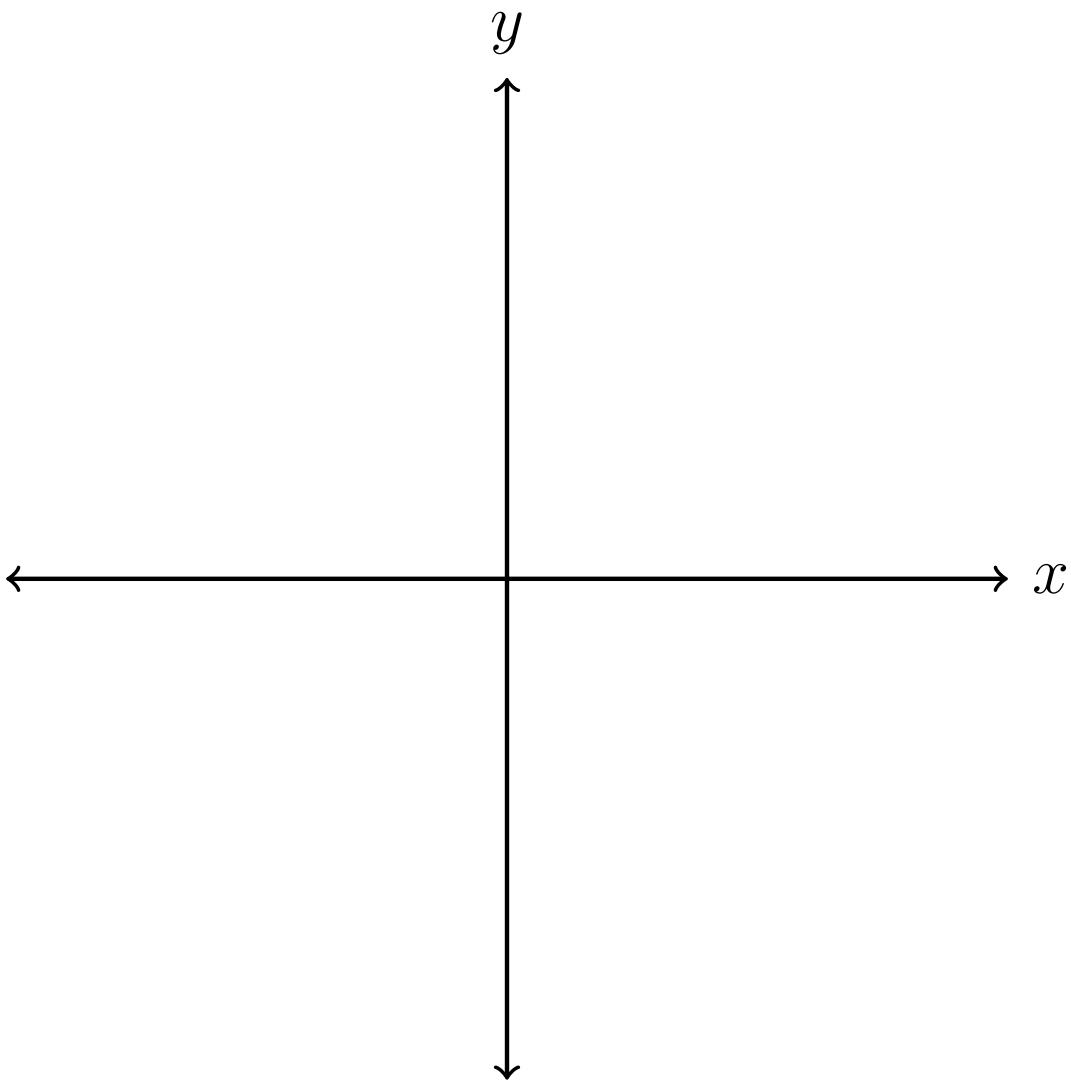
- $f'(x) = 4x^2(x-3)$
- Critical points at $x = 0$ and $x = 3$, where $f'(x) = 0$.
- $f'(x) > 0$ on $(3, \infty)$.
- $f'(x) < 0$ on $(-\infty, 0) \cup (0, 3)$.
- Local minimum at $x = 3$.
- No local maximum. *(Note: The derivative equals* 0 *at* $x = 0$*, but does not change sign, so it is neither a max nor a min!)*

Second Derivative (Concavity & Inflection):

- $f''(x) = 12x(x-2)$
- $f''(x) > 0$ on $(-\infty, 0) \cup (2, \infty)$.
- $f''(x) < 0$ on $(0, 2)$.
- Points of inflection at $x = 0$ and $x = 2$.

Points on the Graph (Function Values):

- $f(0) = 10$ (y-intercept and point of inflection).
- $f(2) = -6$ (point of inflection).
- $f(3) = -17$ (local minimum).


y
x

6. Consider the radical function $f(x) = \sqrt[3]{x} + 2$. Sketch the graph using the provided information.

Domain and Asymptotes:

- The domain of $f(x)$ is $(-\infty, \infty)$.
- No vertical or horizontal asymptotes.

First Derivative (Critical Points & Increasing/Decreasing):

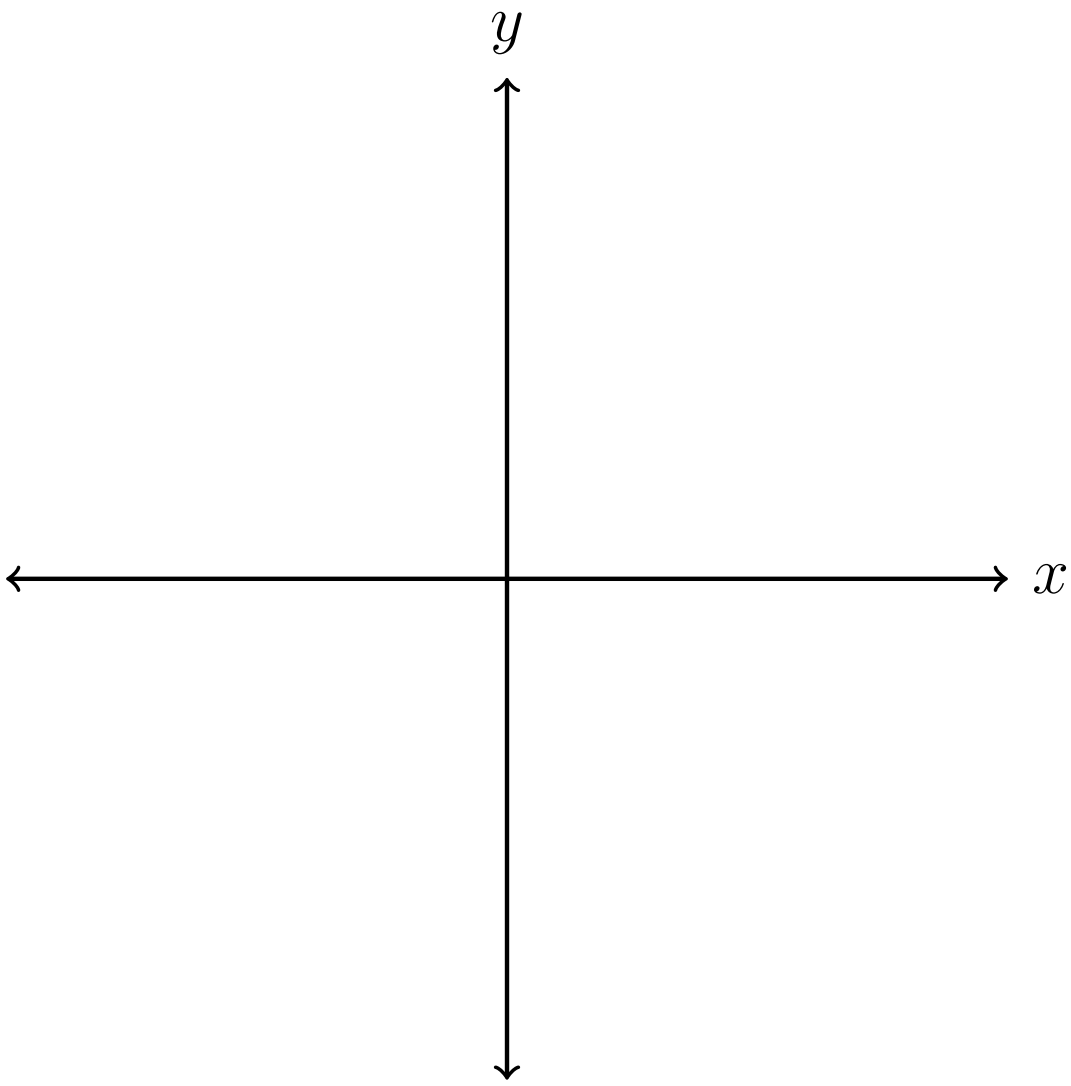
- $f'(x) = \dfrac{1}{3x^{2/3}}$
- Critical point at $x = 0$, where $f'(0)$ does not exist.
- $f'(x) > 0$ on $(-\infty, 0) \cup (0, \infty)$.
- The function is strictly increasing.
- No local extrema.

Second Derivative (Concavity & Inflection):

- $f''(x) = \dfrac{-2}{9x^{5/3}}$
- $f''(x) > 0$ on $(-\infty, 0)$.
- $f''(x) < 0$ on $(0, \infty)$.
- Point of inflection at $x = 0$.

Points on the Graph (Function Values):

- $f(0) = 2$ (point of inflection & y-intercept).
- $f(1) = 3$.
- $f(-1) = 1$.
- $f(-8) = 0$ (x-intercept).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